

Validation of a portable force platform (PhyPlate) for biomechanical jump analysis

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Abstract: In sports sciences, Ground Reaction Forces (GRF) data allow practitioners and scientists to analyze kinetics and kinematics during various types of activities such as counter-movement jumps (CMJ) or squat jumps (SJ). The GRF data are usually obtained using force platforms, however, these devices have several drawbacks (cost, portability, ease of use, etc.), which limit their use on the field. Thus, the aim of this study was to assess the validity of a portable force platform measuring vertical GRF (PhyPlate) by comparing its data to a state-of-the-art force platform (AMTI) during static protocol and during CMJs. Results indicate that the PhyPlate is a very reliable tool to measure vertical GRF, showing 0.04% bias with the gold standard force platform (AMTI) during static protocol and RMSE of less than 0.01% during CMJs.

1. Introduction

Ground Reaction Forces (GRF) are widely studied in sports sciences to evaluate performance and injury risk. These data can be obtained using force platforms, which are the gold standard devices for GRF measurement. Force platforms are composed of force sensors and are used to measure the forces they are subjected to. They are typically used during Counter-Movement Jumps (CMJ), Squat Jumps (SJ) or Drop Jumps (DJ) to assess athlete explosiveness [1], strength [2], or to monitor fatigue [3] using parameters such as jump height or reactive strength index (RSI). Moreover, the vertical component of GRF allows for the assessment of jump performances through force-time profile characteristics [4], [5].

However, these devices are expensive, typically mounted only in laboratories, and cannot be easily moved. As a result, they are rarely used in situ, which limits the possibilities of scientists and trainers.

The development of a portable force platform measuring vertical GRF (vGRF) could therefore represent an opportunity for practitioners to assess performance

on-field. The aim of this study was to assess the validity of a newly commercially available force platform (PhyPlate) by comparing its data with reference values obtained with a gold-standard force platform.

2. Methods

2.1. Materials

Two force platforms were used in this study. The PhyPlate (Phyling, France) measured vGRF data at a sampling frequency of 1000 Hz and its size was 30 x 40 cm. This device measured force using four uniaxial force sensors positioned at each corner. The force sensors are composed of strain gauges. In this experiment, two PhyPlates were used (one for each foot). The reference force platform used in this study was the AMTI OR6-6-1000 (AMTI, MA, USA). It measured 120 x 60 cm and its data acquisition sampling frequency was also set to 1000Hz. In this study, only the vertical component of this force platform was considered.

2.2. Testing procedure and data analysis

2.2.1. Static protocol

To assess the accuracy of vGRF measurement in a static condition, calibration weights (ADAM Equipment, USA) were placed on the top of the PhyPlate, ranging from 20 to 140 kg. Each weight was gradually placed on the PhyPlate to obtain 13 different levels of weight. For each level, the mean bias between the theoretical weight and the weight measured by the PhyPlate was computed, as well as Bland Altman plots.

2.2.2. Dynamic protocol (CMJ)

PhyPlates were superimposed with the AMTI as described in Figure 1. One participant was instructed to perform 10 CMJ on the superimposed platforms.

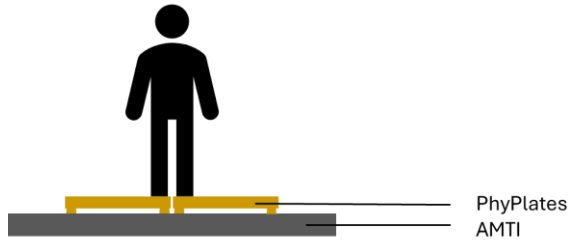


Figure 1. Schematic representation of the experimental setup.

For each CMJ, the coefficient of correlation (R^2), RMSE, NRMSE, and ICC_{3,1} values were calculated between the two measurement devices (AMTI and PhyPlates), taking the whole force-time curve as input. Moreover, CMJ parameters were also calculated using both force platforms (Table 1). Mean values and bias were computed for each parameter.

Table 1. CMJ parameters computed on each measurement devices.

Parameters	Description
JH	Jump Height calculated from jump time [5] (in cm)
Pmax	Maximal Power (in W)
Relative impulse	Relative impulse (in W)
RSI	Reactive Strain Index [1] (Ratio)
CRPD	Mean Concentric Rate of Power Development [6] (Ws/Kg)
ERFD	Mean Eccentric Rate of Force Development [6] (Ns/kg)

3. Results

3.1. Static protocol

The Bland-Altman analysis (Figure 1) revealed a very high degree of agreement between theoretical weights and PhyPlates with a mean relative difference of 0.038%. According to Figure 2, PhyPlates are more accurate when the measured force is above 40 kg.

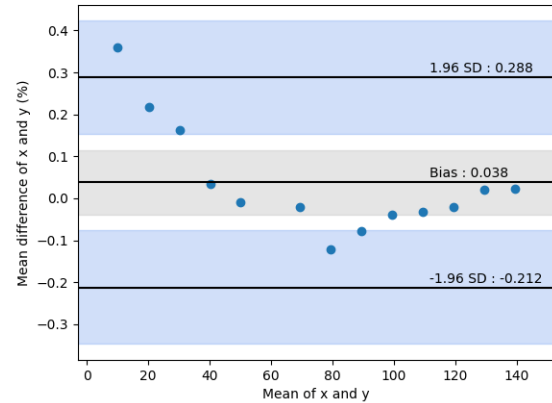


Figure 1 : Bland Altman analysis for static protocol (x corresponds to PhyPlates and y to AMTI)

3.2. Dynamic protocol (CMJ)

Mean values and differences for each CMJ parameter computed are presented in Table 2. A force-time curve comparison between the two measurement devices is presented in Figure 3.

Table 2. CMJ parameters computed by each measurement devices.

Parameters	PhyPlates	AMTI	Diff (%)
JH	35.07 ± 2.01	35.08 ± 1.25	-0.02
Pmax	3070.6 ± 69.1	3068.1 ± 62.0	-0.08
Rel. impulse	2.10 ± 0.05	2.09 ± 0.04	- 0.21
RSI	0.45 ± 0.02	0.45 ± 0.01	-0.08
CRPD	183.39 ± 1.76	183.19 ± 1.84	0.10
ERFD	52.75 ± 6.30	52.66 ± 6.09	0.17

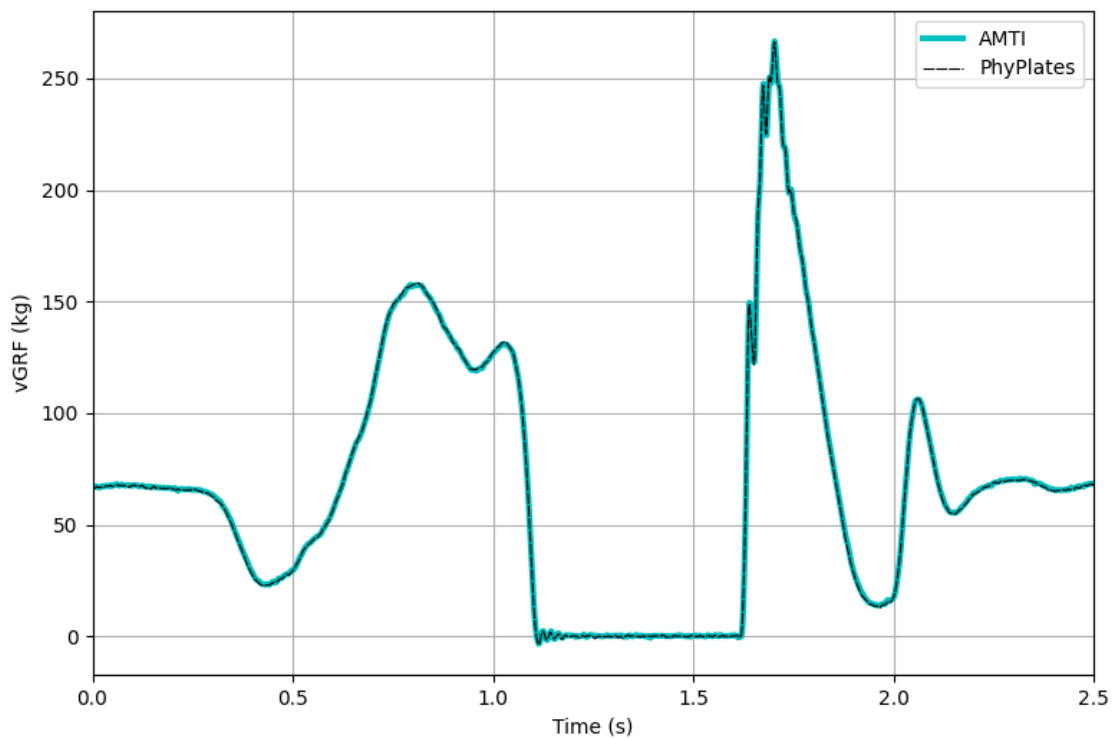


Figure 3. Comparison between the vGRF measured by the AMTI (bold cyan curve) and the vGRF measured by the PhyPlates (black dashed curve) for one CMJ example

Regarding CMJs, NRMSE between the AMTI and the PhyPlate was 0.008% (Table 3).

Table 3. Comparison between the values recorded by the two measurement devices during CMJs.

	Diff. between AMTI and PhyPlate
R²	0.999
RMSE (kg)	0.70 ± 0.03
NRMSE (%)	0.008 ± 0.001
ICC_{3,1} [95% CI]	0.999 [1.0, 1.0]

4. Discussion

PhyPlates show excellent agreement with the reference force platform during both static and dynamic conditions. The very similar values of CMJ parameters computed with the AMTI and the PhyPlates (such as a 0.02% difference in jump height estimation) show that the PhyPlates represent a valid and accurate tool to monitor, assess and measure biomechanical variables during various sport activities (CMJ, SJ, DJ, etc.).

5. Bibliography

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